

Zoe Zhang

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Education

The University of Texas at Dallas

Aug 2023 - May 2027

Bachelor of Science, Computer Science

- **Coursework:** Data Structures, Advanced Algorithms, Database Systems, Operating Systems, Computer Architecture, Data Security, Computer Networks, Software Engineering, Advanced Algorithms Design

Skills

- **Programming Languages:** C++, Java, Python, JavaScript, TypeScript, HTML, CSS, SQL
- **Frameworks & Libraries:** Flask, FastAPI, React, PyTorch, ResNet8, AWS SDK, Android SDK, Paho MQTT
- **Security & Cloud:** AWS, AWS IoT Core, Amazon S3, AWS SDK, mTLS, Android Keystore, X.509 certificates, MQTT, TLS 1.2+

Experience

Infosys | *Software Engineering Intern*

Jun 2026 - Present

Security Lead - Secure Sentry Robot (Android/Temi robot, AWS IoT cloud)

- Own the device-to-cloud security layer for a 6-person autonomous-robot project, designing mutual TLS (mTLS) authentication between an Android-based Temi robot and AWS IoT Core over MQTT.
- Built a hardware-backed key-management module in Java that imports X.509 device credentials into the Android Keystore (StrongBox/TEE) as non-exportable keys, removing private keys from on-device plaintext storage.
- Designed and validated a test PKI (EC P-256 root CA, server/device certificates) with OpenSSL, demonstrating a successful mutual-TLS handshake (TLS 1.2) and verifying clients without CA-signed certificates.

University of Mississippi Medical Center | *Data Science Research & Lab Assistant*

May 2025 - Aug 2025

- Led daily lab sessions to support instruction in data science applications, core concepts in Python programming, and software engineering fundamentals including statistical modeling, machine learning, and natural language processing.
- Tutored and assisted students in Python programming and data analysis by guiding them through real-world addiction-related datasets (national health surveys, electronic health records, social media data), applying strong problem-solving skills and emphasizing data handling best practices and exploratory analysis.

UTD University Recreation | *Graphic Designer*

Sep 2024 - Jun 2025

- Collaborated with the marketing team to design and optimize flyers, banners, and digital graphics using Adobe Illustrator, Photoshop, and web technologies, translating ideas into engaging visuals that promoted campus events, reinforced UTD's brand identity, and met project deadlines.

Projects

Ivory: Product Recognition App

May 2025 - Aug 2025

- Designed and developed a full-stack Flask application that uses computer vision (ResNet18) to identify and compare uploaded product images against an inventory database. The inventory system can view product details, add new items with metadata, and make them searchable in future uploads.
- Structured the project with a maintainable backend (Flask, PyTorch, JSON-based storage) and frontend templates (HTML/CSS), enabling scalability for future search, analytics, and sales features.

Airline Passenger Service Web App (Frontend)

Jan 2025

- Developed a web platform that connects users directly to airline service and management systems, enabling airline passengers to order food, request in-flight items, and call for flight attendants directly from their device, enhancing the overall customer experience and streamlining crew service operations.
- Utilized Typescript and CSS to implement asynchronous data handling and UI optimization, ensuring real-time communication between passengers and flight attendants with minimal latency.

Extracurricular

NIH Data Science for Addiction Research | *Co-Author*

Jul 2025 - Aug 2025

- Conducted a systematic literature review of studies applying data science and machine learning methods (NLP, random forest) to research on child maltreatment and substance use disorders in youth populations.
- Synthesized findings on model performance, predictive factors, and methodological gaps, highlighting future opportunities for AI-driven approaches in public health research.

Association for Computing Machinery

Aug 2023 - Present

Fillipino Student Association | *Graphics Chair*

Aug 2023 - Present